الجامعة السعودية الإلكترونية
SAUDI ELECTRONIC UNIVERSITY
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College of Computing and Informatics

Fundamentals of Database

IT403

Project – Mid Report

Deadline: Monday 09/08/2026 @ 23:59

[Total Mark is 20]

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Instructions:

- You must submit two separate copies **(one Word file and one PDF file)** using the Assignment Template on Blackboard via the allocated folder. These files **must not be in compressed format**.
- It is your responsibility to check and make sure that you have uploaded both the correct files.
- Zero mark will be given if you try to bypass the SafeAssign (e.g. misspell words, remove spaces between words, hide characters, use different character sets, convert text into image or languages other than English or any kind of manipulation).
- Email submission will not be accepted.
- You are advised to make your work clear and well-presented. This includes filling your information on the cover page.
- You must use this template, failing which will result in zero mark.
- You MUST show all your work, and text must not be converted into an image, unless specified otherwise by the question.
- Late submission will result in ZERO mark.
- The work should be your own, copying from students or other resources will result in ZERO mark.
- Use **Times New Roman** font for all your answers.

Mid Report

1. Work completed this period

During this reporting period, the team successfully completed several important milestones required for the project

The first step involved analyzing the project requirements provided in the assignment. We carefully studied the clinic scenario and identified the main entities required for the database. After discussing different design possibilities, we selected the most suitable entities, including Patient, Employee, Doctor, Receptionist, Appointment, Treatment, Medicine, and Payment

After identifying the entities, we designed the Entity Relationship (ER) diagram and then extended it into an Enhanced Entity Relationship (EER) diagram by introducing specialization. We defined all primary keys, foreign keys, relationships, and cardinalities while ensuring that the database structure follows normalization principles

Once the database design was completed, the implementation phase started. We created the SmartClinic database using MySQL and implemented all required tables with appropriate data types and constraints such as PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE, and AUTO_INCREMENT

The team also inserted sample records into each table to simulate a real clinic environment. This data allowed us to verify that the relationships between tables were functioning correctly and that all constraints were enforced properly

To support collaboration, a GitHub repository was created where the project files were uploaded and organized. This allowed the team members to keep track of their work and maintain version control throughout the project

Overall, the first phase established a complete database structure that will be used for implementing the remaining SQL operations in the next stage

2. One key decision has been made during this period

One of the most important decisions made by the team was choosing to use an Employee superclass with Doctor and Receptionist as specialized subclasses

Initially, we considered creating completely separate tables for doctors and receptionists, each containing personal information such as name, phone number, email, and salary. However, this approach would have duplicated many common attributes

Instead, we adopted the specialization feature of the EER model. The Employee table stores the common employee information, while Doctor and Receptionist contain only their specialized attributes

This design offers several advantages:

- It minimizes data redundancy
- It improves database normalization
- It simplifies maintenance
- It allows new employee types to be added in the future without redesigning the database
- It follows good database design principles taught during the course

Although this approach required additional foreign key relationships, the benefits outweighed the extra complexity

3. One problem encountered

During the implementation process, the team encountered several challenges

The main difficulty was ensuring that all relationships between the tables were correctly defined while maintaining proper normalization. Since many tables depend on one another through foreign keys, the order of table creation and data insertion became very important

For example, the Employee table had to be created before the Doctor and Receptionist tables because both depend on Employee. Similarly, Patient and Doctor records had to exist before Appointment records could be inserted successfully

Another challenge involved verifying that every relationship represented the clinic's business rules correctly. We carefully reviewed the ER/EER diagrams and tested the database after each implementation step to ensure that no constraint violations occurred

To solve these issues, we divided the implementation into smaller stages, tested every table individually, and verified foreign key relationships before continuing to the next step. This approach reduced errors and ensured that the database behaved as expected

4. Plan to do next

During the next phase of the project, the team plans to complete all remaining project requirements

The first task will be implementing the required SQL operations, including multiple SELECT statements, JOIN queries, nested queries, aggregate functions with GROUP BY, UPDATE statements, and DELETE statements

After that, we will create a VIEW that simplifies frequently used queries and implement a TRIGGER to improve data integrity

Once all SQL operations have been completed, we will perform extensive testing to verify that every query produces the expected results. We will also capture screenshots of both the SQL code and its output as required by the project instructions

Finally, we will prepare the reflection section, organize the project report, review the GitHub repository, verify that all required files are included, and ensure that the final submission satisfies every requirement listed in the project specification

5. Each group member input/participation

The project work was divided among the team members according to the assignment tasks

Mohammed Alsaleem (Task 1 – Database Design)

Responsible for analyzing the system requirements, identifying the entities, designing the ER and EER diagrams, defining relationships, cardinalities, primary keys, foreign keys, specialization, and documenting the design assumptions

Abdulrahman Alabdulwahid (Task 2 – Database Implementation)

Responsible for implementing the database in MySQL, creating all database tables, defining constraints, and inserting sample records into every table while verifying database integrity

Mohammed Alhumud (Task 3 – SQL Operations)

Responsible for writing the required SQL queries, including SELECT statements, JOIN operations, nested queries, aggregate functions, UPDATE and DELETE statements, and creating the VIEW and TRIGGER

Adel Almalki (Task 4 – Reflection)

Responsible for preparing the project reflection, documenting the team's experience, discussing the challenges encountered, explaining the design decisions, and reviewing the final report before submission